

CURRICULUM BOOK

UNDERGRADUATE PROGRAM: BACHELOR OF ENGINEERING

COMPUTER ENGINEERING

Effective from Academic Year: 2026-27

Status: Autonomous College Syllabus Handbook

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Principal

ACADEMIC PREAMBLE

As an autonomous engineering college affiliated with the University of Mumbai, Fr. Conceicao Rodrigues College of Engineering is dedicated to engineering excellent learners, innovators, and future technical leaders. Autonomy enables the institution to respond to technological shifts and design responsive curricula that balance theoretical foundations with experiential design, modern tools, and professional ethics.

This curriculum structure represents a unified framework curated in collaboration with academic advisors and leading technical firms. The curriculum emphasizes robust mathematical models, project-based validation, and outcome-oriented professional practice.

Vision of the Department

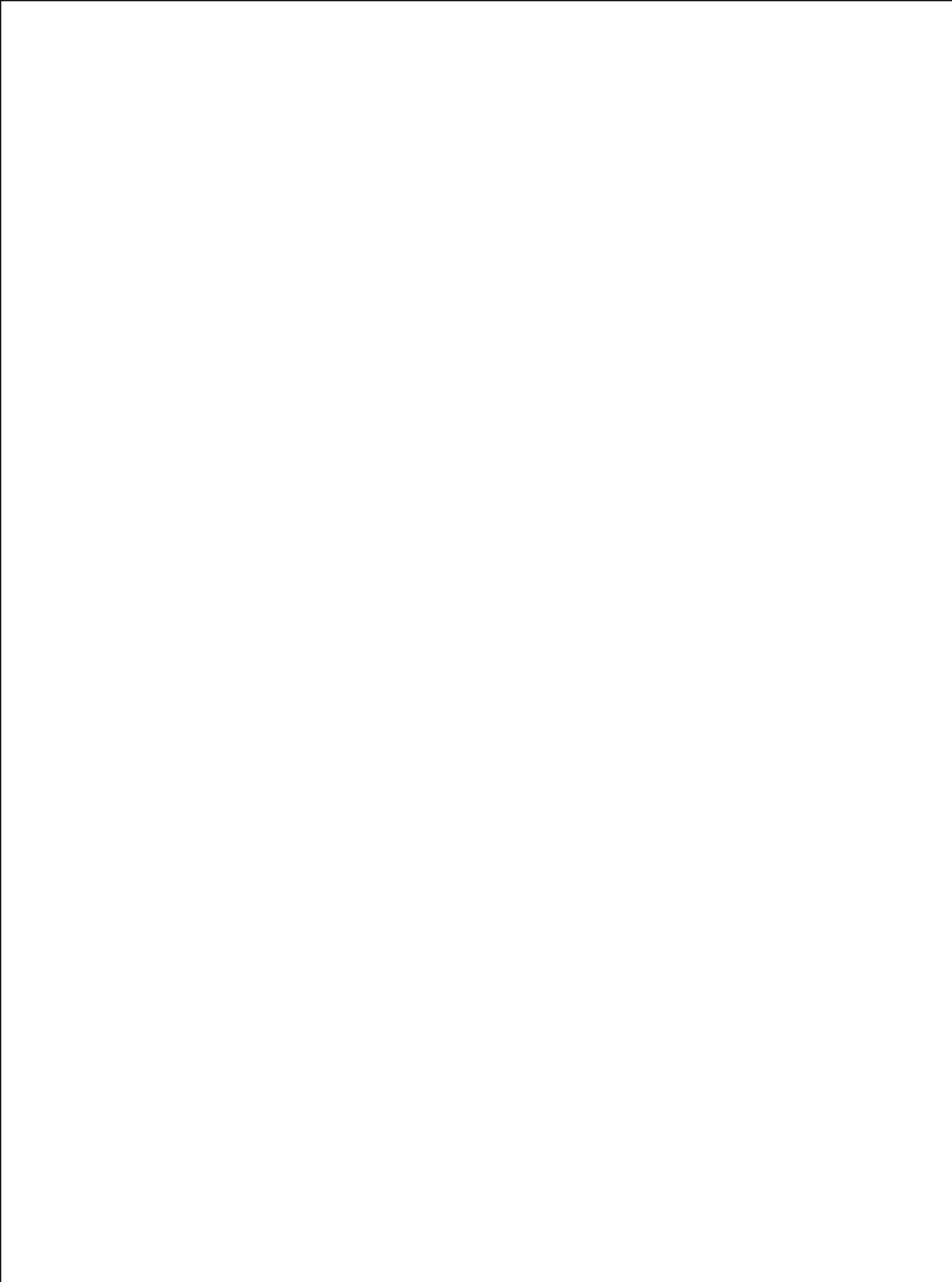
To be a leading center of excellence in engineering education and research, fostering industry-ready technical innovators, ethically sound professionals, and empathetic citizens.

Mission of the Department

- Deliver standard-aligned engineering education using student-centric pedagogy and laboratory environments.
- Foster research, technical incubation, and industry-institution partnerships.
- Cultivate life-long learning habits, teamwork capabilities, and professional ethical standards.

CURRICULUM ARCHITECTURE AND NEP ALIGNMENT**Course Nomenclature Table**

Category	Course Category Name	NEP-2020 Credit Framework Description
BS	Basic Sciences	Fundamental disciplines including Physics, Chemistry, and Advanced Calculus.
ES	Engineering Sciences	Interdisciplinary foundations such as Computing, Graphics, and Electrical Systems.
PC	Professional Core	Core discipline-specific subjects, advanced concepts, and matching practicals.
PE	Professional Electives	Branching streams for specialized expertise such as AI/ML, Security, and IoT.
OE	Open Electives	Multi-disciplinary courses offered across departments for wider learning.
HS	Humanities and Social Sciences	Professional Communication, Business Ethics, IPR, and Management.



SEMESTERWISE CURRICULUM STRUCTURE

Computer Engineering Program: Semester 3

Course Code	Course Name	Teaching Scheme (Contact Hours/Week)			Credits Assigned				Examination Scheme						
		L	T	P	L	T	P	Total	Theory Courses				Practical / Lab		Total Marks
									ISE 1	ISE 2	MSE	ESE	ISE 1	ISE 2	
CSC301	Data Structures and Algorithms	3	1	0	3	1	0	4	20	20	-	60	-	-	100
CSL301	Data Structures Laboratory	0	0	2	0	0	1	1	20	20	-	-	25	25	100
CSC302	Database Management Systems	3	0	0	3	0	0	3	20	20	-	60	-	-	100
CSL302	Database Systems Laboratory	0	0	2	0	0	1	1	20	20	-	-	25	25	100
CSC303	Computer Networks	3	1	0	3	1	0	4	20	20	-	60	-	-	100
CSC304	Operating Systems	3	0	0	3	0	0	3	20	20	-	60	-	-	100
CSE305	Professional Elective: Artificial Intelligence	3	0	0	3	0	0	3	20	20	-	60	-	-	100
CSP306	Mini Project I	0	0	4	0	0	2	2	20	20	-	-	25	25	100
Total		15	2	8	21				-						

Abbreviations: L: Lecture; T: Tutorial; P: Practical/Lab; ISE: In-Semester Evaluation; MSE: Mid-Semester Examination; ESE: End Semester Examination.

Regulatory Ordinance Code: R-2026-COMP-III

SEMESTERWISE CURRICULUM STRUCTURE

Computer Engineering Program: Semester 4

Course Code	Course Name	Teaching Scheme (Contact Hours/Week)			Credits Assigned				Examination Scheme						
		L	T	P	L	T	P	Total	Theory Courses				Practical / Lab		Total Marks
									ISE 1	ISE 2	MSE	ESE	ISE 1	ISE 2	
CSC401	Design and Analysis of Algorithms	3	1	0	3	1	0	4	20	20	-	60	-	-	100
CSL401	Algorithm Design Laboratory	0	0	2	0	0	1	1	20	20	-	-	25	25	100
CSC402	Software Engineering	3	0	0	3	0	0	3	20	20	-	60	-	-	100
CSC403	Computer Architecture	3	1	0	3	1	0	4	20	20	-	60	-	-	100
CSE404	Professional Elective: Cyber Security	3	0	0	3	0	0	3	20	20	-	60	-	-	100
CSO405	Open Elective: Data Visualization	2	0	2	2	0	1	3	20	20	-	60	25	25	150
CSP406	Community Innovation Project	0	0	4	0	0	2	2	20	20	-	-	25	25	100
Total		14	2	8	20				-						

Abbreviations: L: Lecture; T: Tutorial; P: Practical/Lab; ISE: In-Semester Evaluation; MSE: Mid-Semester Examination; ESE: End Semester Examination.

Regulatory Ordinance Code: R-2026-COMP-IV

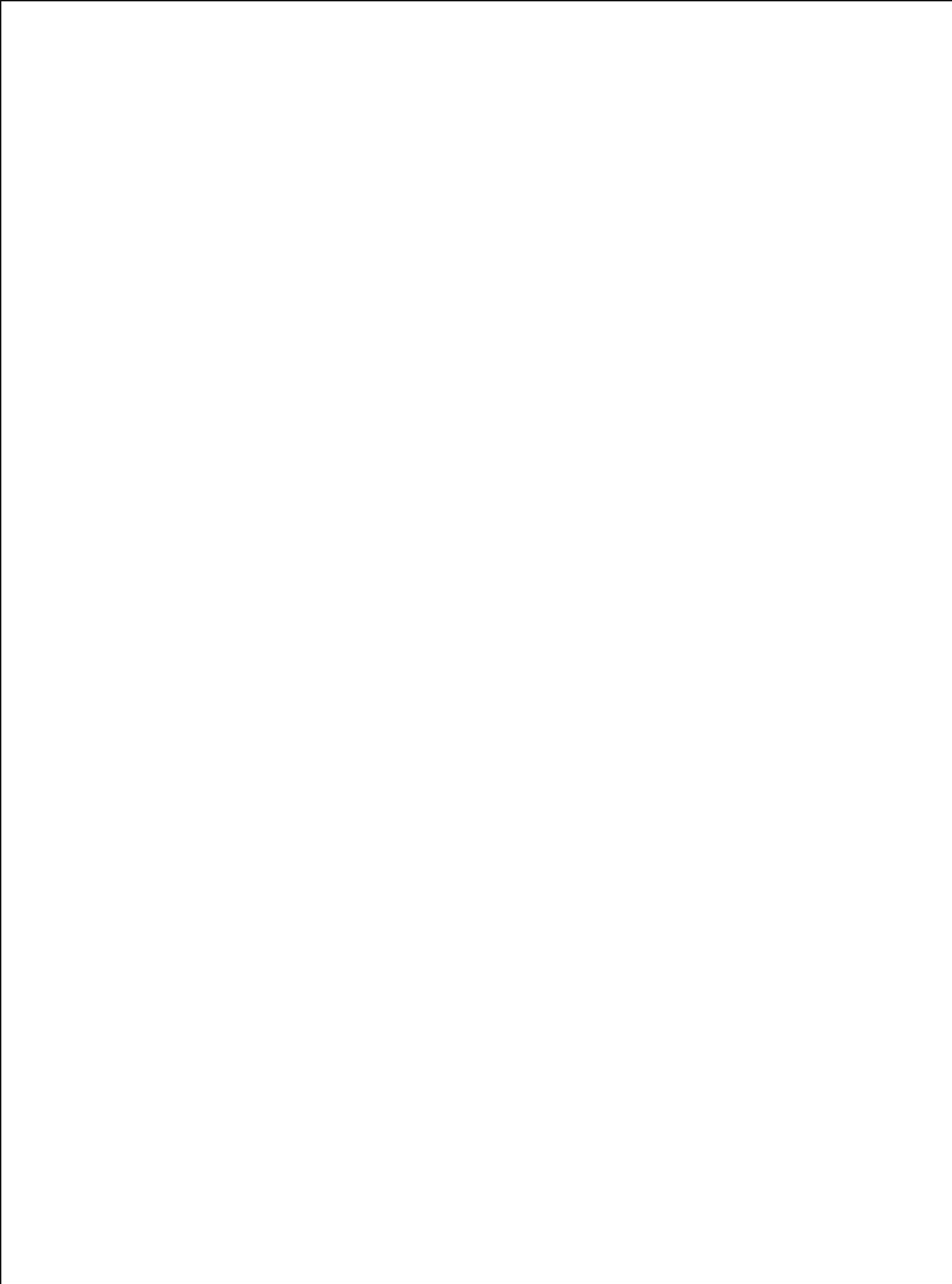
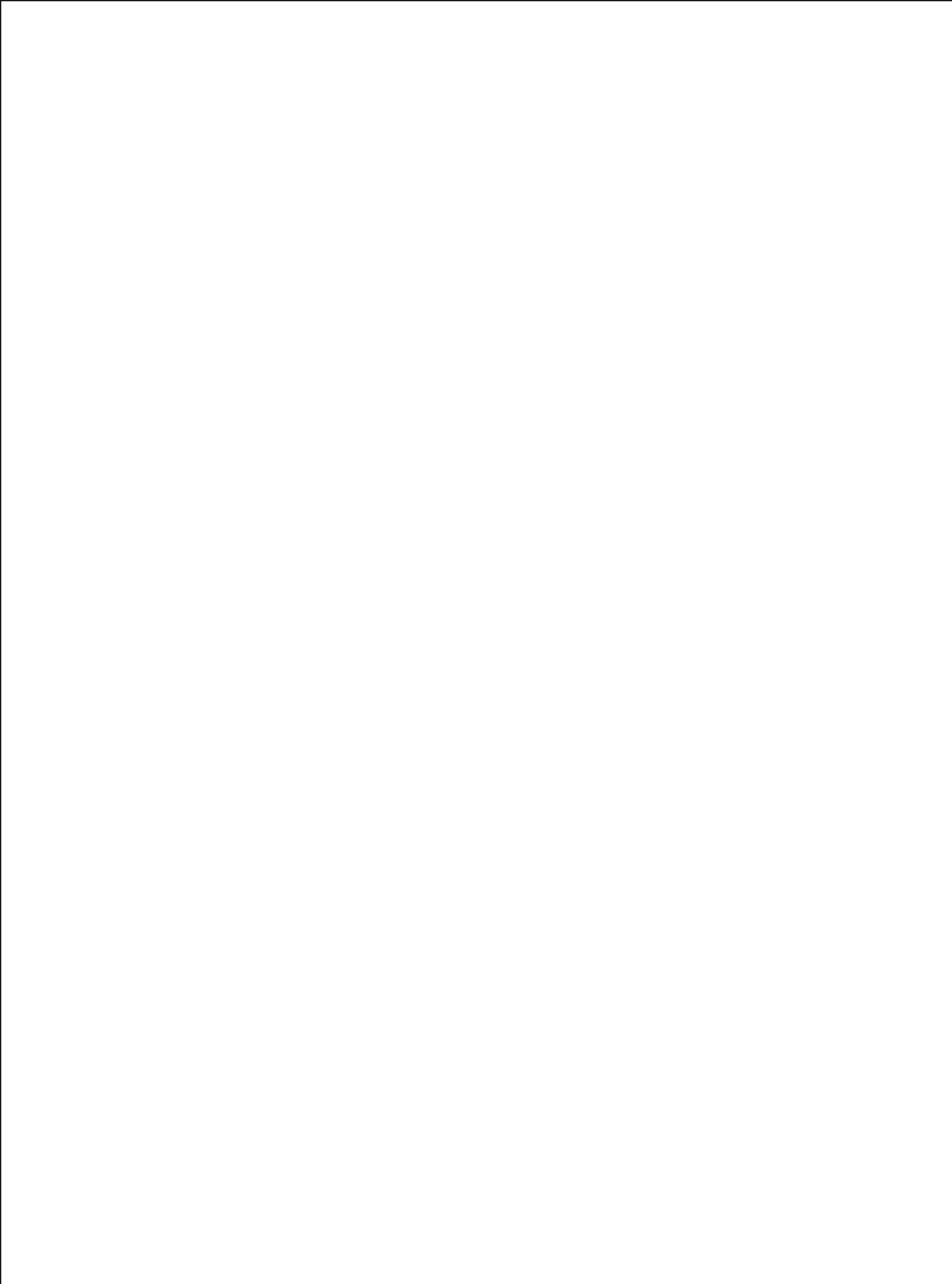


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Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		3	1	--	--	3	1	--	4
CSC301	Data Structures and Algorithms	Examination Scheme							
			ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	60	100		

Pre-requisite Course Codes	Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify data structures and algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain data structures and algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply data structures and algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze data structures and algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design data structures and algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Course Assessment:**Theory:**

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Data Structures and Algorithms: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Data Structures and Algorithms. Pearson, 1st, 2024.

Department Faculty Board. Data Structures and Algorithms Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



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Course Content (Practical Only)

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		--	--	2	--	--	--	1	1
		Examination Scheme							
CSL301	Data Structures Laboratory		ISE1	MSE	ISE2	ESE	Total		
		Theory	--	--	--	--	--		
		Lab	25	--	25	--	50		

Pre-requisite Course Codes	Nil
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify data structures laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain data structures laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply data structures laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze data structures laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design data structures laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Course Assessment: (Lab)

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

Laboratory Rubric (25 marks): Continuous laboratory evaluation using predefined rubrics and viva voce.

To be Taught in laboratory			
	Topics wise List of Experiments with relevant topic	Ref.	COs
1	Data Structures Laboratory Experiment 1: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
2	Data Structures Laboratory Experiment 2: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
3	Data Structures Laboratory Experiment 3: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
4	Data Structures Laboratory Experiment 4: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
5	Data Structures Laboratory Experiment 5: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
6	Data Structures Laboratory Experiment 6: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--

Recommended Books:

A. Kulkarni, M. Rao. Data Structures Laboratory: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Data Structures Laboratory. Pearson, 1st, 2024.

Department Faculty Board. Data Structures Laboratory Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



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Course Content

Course Code CSC302	Course Name Database Management Systems	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		3	--	--	--	3	--	--	3
		Examination Scheme							
			ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	60	100		

Pre-requisite Course Codes	Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify database management systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain database management systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply database management systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze database management systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design database management systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Course Assessment:**Theory:**

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Database Management Systems: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Database Management Systems. Pearson, 1st, 2024.

Department Faculty Board. Database Management Systems Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



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Course Content (Practical Only)

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		--	--	2	--	--	--	1	1
CSL302	Database Systems Laboratory	Examination Scheme							
			ISE1	MSE	ISE2	ESE	Total		
		Theory	--	--	--	--	--		
		Lab	25	--	25	--	50		

Pre-requisite Course Codes	Nil
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify database systems laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain database systems laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply database systems laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze database systems laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design database systems laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Course Assessment: (Lab)

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

Laboratory Rubric (25 marks): Continuous laboratory evaluation using predefined rubrics and viva voce.

To be Taught in laboratory			
	Topics wise List of Experiments with relevant topic	Ref.	COs
1	Database Systems Laboratory Experiment 1: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
2	Database Systems Laboratory Experiment 2: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
3	Database Systems Laboratory Experiment 3: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
4	Database Systems Laboratory Experiment 4: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
5	Database Systems Laboratory Experiment 5: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
6	Database Systems Laboratory Experiment 6: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--

Recommended Books:

A. Kulkarni, M. Rao. Database Systems Laboratory: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Database Systems Laboratory. Pearson, 1st, 2024.

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CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



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Course Content

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		3	1	--	--	3	1	--	4
CSC303	Computer Networks	Examination Scheme							
			ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	60	100		

Pre-requisite Course Codes	Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify computer networks concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain computer networks concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply computer networks concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze computer networks concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design computer networks concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total				46

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total				46

Course Assessment:**Theory:**

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Computer Networks: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Computer Networks. Pearson, 1st, 2024.

Department Faculty Board. Computer Networks Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		3	--	--	--	3	--	--	3
CSC304	Operating Systems	Examination Scheme							
			ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	60	100		

Pre-requisite Course Codes	Nil
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify operating systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain operating systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply operating systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze operating systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design operating systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total				46

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total				46

Course Assessment:**Theory:**

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Operating Systems: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Operating Systems. Pearson, 1st, 2024.

Department Faculty Board. Operating Systems Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



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Course Content

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		3	--	--	--	3	--	--	3
CSE305	Professional Elective: Artificial Intelligence	Examination Scheme							
			ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	60	100		

Pre-requisite Course Codes	Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify professional elective: artificial intelligence concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain professional elective: artificial intelligence concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply professional elective: artificial intelligence concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze professional elective: artificial intelligence concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design professional elective: artificial intelligence concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total				46

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Course Assessment:**Theory:**

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Professional Elective: Artificial Intelligence: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Professional Elective: Artificial Intelligence. Pearson, 1st, 2024.

Department Faculty Board. Professional Elective: Artificial Intelligence Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



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Course Content (includes Practical)

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		--	--	4	--	--	--	2	2
		Examination Scheme							
CSP306	Mini Project I		ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	--	50		
		Lab	25	--	25	--	50		

Pre-requisite Course Codes	Nil
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify mini project i concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain mini project i concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply mini project i concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze mini project i concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design mini project i concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			0	

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			0	

Course Assessment:**Theory:**

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

Laboratory Rubric (25 marks): Continuous laboratory evaluation using predefined rubrics and viva voce.

To be Taught in laboratory			
	Topics	Ref.	COs
1	Mini Project I Experiment 1: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
2	Mini Project I Experiment 2: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
3	Mini Project I Experiment 3: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
4	Mini Project I Experiment 4: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
5	Mini Project I Experiment 5: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
6	Mini Project I Experiment 6: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--

Recommended Books:

A. Kulkarni, M. Rao. Mini Project I: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Mini Project I. Pearson, 1st, 2024.

Department Faculty Board. Mini Project I Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



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Course Content

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		3	1	--	--	3	1	--	4
CSC401	Design and Analysis of Algorithms	Examination Scheme							
			ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	60	100		

Pre-requisite Course Codes	Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify design and analysis of algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain design and analysis of algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply design and analysis of algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze design and analysis of algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design design and analysis of algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total				46

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Course Assessment:**Theory:**

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Design and Analysis of Algorithms: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Design and Analysis of Algorithms. Pearson, 1st, 2024.

Department Faculty Board. Design and Analysis of Algorithms Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



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Course Content (Practical Only)

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		--	--	2	--	--	--	1	1
		Examination Scheme							
CSL401	Algorithm Design Laboratory		ISE1	MSE	ISE2	ESE	Total		
		Theory	--	--	--	--	--		
		Lab	25	--	25	--	50		

Pre-requisite Course Codes	Nil
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify algorithm design laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain algorithm design laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply algorithm design laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze algorithm design laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design algorithm design laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Course Assessment: (Lab)

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

Laboratory Rubric (25 marks): Continuous laboratory evaluation using predefined rubrics and viva voce.

To be Taught in laboratory			
	Topics wise List of Experiments with relevant topic	Ref.	COs
1	Algorithm Design Laboratory Experiment 1: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
2	Algorithm Design Laboratory Experiment 2: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
3	Algorithm Design Laboratory Experiment 3: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
4	Algorithm Design Laboratory Experiment 4: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
5	Algorithm Design Laboratory Experiment 5: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
6	Algorithm Design Laboratory Experiment 6: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--

Recommended Books:

A. Kulkarni, M. Rao. Algorithm Design Laboratory: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Algorithm Design Laboratory. Pearson, 1st, 2024.

Department Faculty Board. Algorithm Design Laboratory Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



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Course Content

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		3	--	--	--	3	--	--	3
		Examination Scheme							
			ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	60	100		

Pre-requisite Course Codes	Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify software engineering concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain software engineering concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply software engineering concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze software engineering concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design software engineering concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total				46

Course Assessment:**Theory:**

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Software Engineering: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Software Engineering. Pearson, 1st, 2024.

Department Faculty Board. Software Engineering Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		3	1	--	--	3	1	--	4
CSC403	Computer Architecture	Examination Scheme							
			ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	60	100		

Pre-requisite Course Codes	Nil
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify computer architecture concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain computer architecture concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply computer architecture concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze computer architecture concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design computer architecture concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total				46

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total				46

Course Assessment:**Theory:**

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Computer Architecture: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Computer Architecture. Pearson, 1st, 2024.

Department Faculty Board. Computer Architecture Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



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Course Content

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		3	--	--	--	3	--	--	3
		Examination Scheme							
CSE404	Professional Elective: Cyber Security		ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	60	100		

Pre-requisite Course Codes	Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify professional elective: cyber security concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain professional elective: cyber security concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply professional elective: cyber security concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze professional elective: cyber security concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design professional elective: cyber security concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total				46

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Course Assessment:**Theory:**

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Professional Elective: Cyber Security: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Professional Elective: Cyber Security. Pearson, 1st, 2024.

Department Faculty Board. Professional Elective: Cyber Security Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

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Course Content (includes Practical)

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		2	--	2	--	2	--	1	3
CSO405	Open Elective: Data Visualization	Examination Scheme							
			ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	60	100		
		Lab	25	--	25	--	50		

Pre-requisite Course Codes	Nil
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify open elective: data visualization concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain open elective: data visualization concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply open elective: data visualization concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze open elective: data visualization concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design open elective: data visualization concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Course Assessment:**Theory:**

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

To be Taught in laboratory			
	Topics	Ref.	COs
1	Open Elective: Data Visualization Experiment 1: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
2	Open Elective: Data Visualization Experiment 2: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
3	Open Elective: Data Visualization Experiment 3: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
4	Open Elective: Data Visualization Experiment 4: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
5	Open Elective: Data Visualization Experiment 5: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
6	Open Elective: Data Visualization Experiment 6: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--

Recommended Books:

A. Kulkarni, M. Rao. Open Elective: Data Visualization: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Open Elective: Data Visualization. Pearson, 1st, 2024.

Department Faculty Board. Open Elective: Data Visualization Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



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Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

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Course Content (includes Practical)

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		--	--	4	--	--	--	2	2
		Examination Scheme							
CSP406	Community Innovation Project		ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	--	50		
		Lab	25	--	25	--	50		

Pre-requisite Course Codes	Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify community innovation project concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain community innovation project concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply community innovation project concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze community innovation project concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design community innovation project concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			0	

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			0	

Course Assessment:**Theory:**

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

Laboratory Rubric (25 marks): Continuous laboratory evaluation using predefined rubrics and viva voce.

To be Taught in laboratory			
	Topics	Ref.	COs
1	Community Innovation Project Experiment 1: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
2	Community Innovation Project Experiment 2: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
3	Community Innovation Project Experiment 3: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
4	Community Innovation Project Experiment 4: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
5	Community Innovation Project Experiment 5: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
6	Community Innovation Project Experiment 6: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--

Recommended Books:

A. Kulkarni, M. Rao. Community Innovation Project: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Community Innovation Project. Pearson, 1st, 2024.

Department Faculty Board. Community Innovation Project Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3

ANNEXURE A: ACADEMIC POLICIES

1. Evaluation Framework

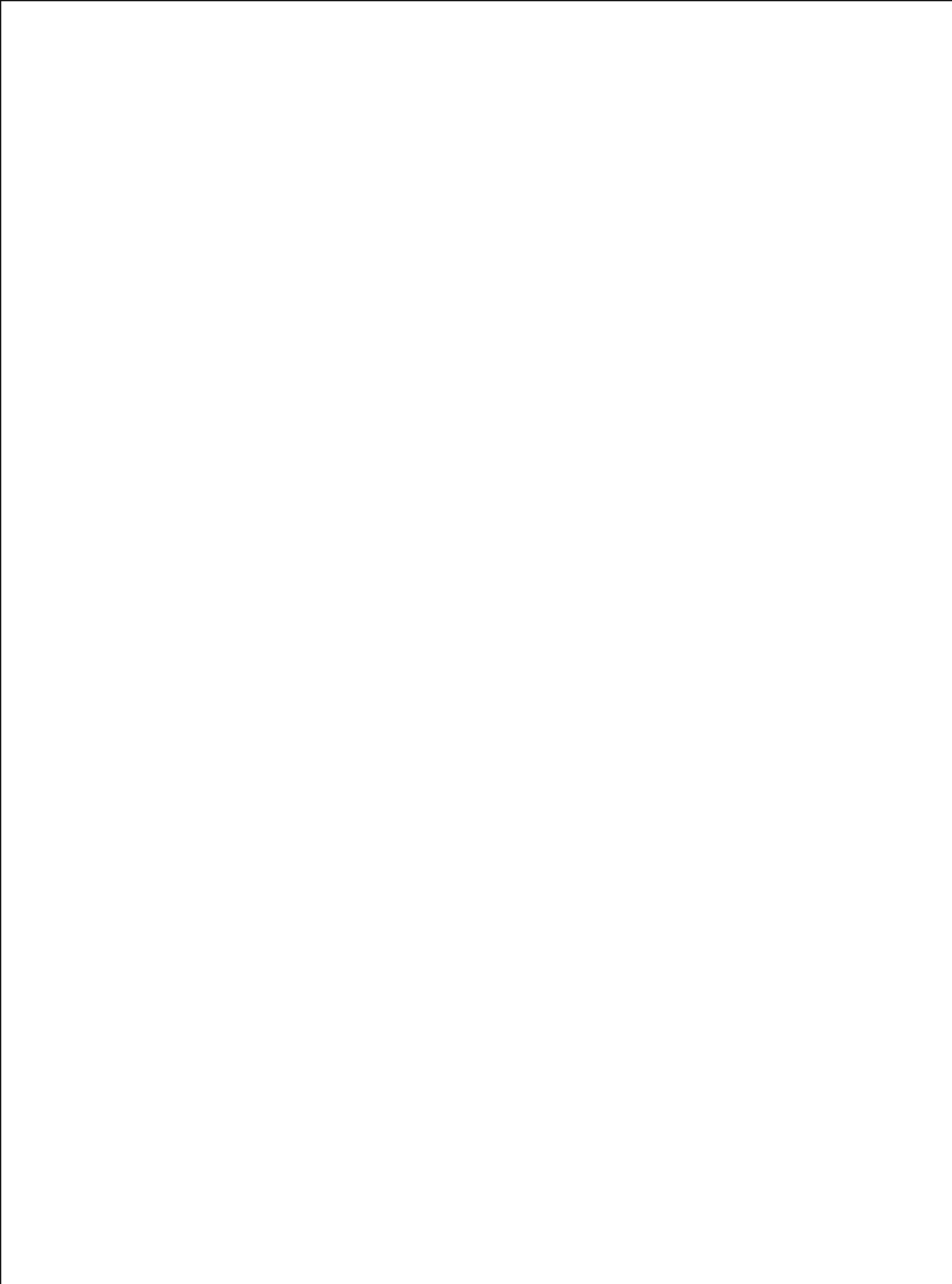
The academic evaluation uses Continuous Internal Evaluation and End Semester Examination components. Students must secure the minimum passing threshold as defined by autonomous college regulations.

2. Grading System

Under the autonomous framework, grading follows a standard relative system converted to a 10-point CGPA scale.

3. Syllabus Revision Protocol

Major syllabus revisions occur periodically to absorb technological shifts. Minor corrections and elective introductions may occur annually with Board of Studies recommendations and Academic Council approval.



REVIEWER COMMENTS SECTION

Course	Section	Comment	Status
CSC301	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved
CSC301	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSL301	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Open
CSL301	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSC302	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved
CSC302	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSL302	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Open
CSL302	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSC303	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved
CSC303	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSC304	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Open
CSC304	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSE305	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved
CSE305	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSP306	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Open
CSP306	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSC401	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved

Course	Section	Comment	Status
CSC401	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSL401	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Open
CSL401	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSC402	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved
CSC402	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSC403	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Open
CSC403	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSE404	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved
CSE404	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSO405	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Open
CSO405	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSP406	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved
CSP406	Modules	Check whether long module blocks remain intact across print page boundaries.	Open